

Student Name: .....



## 2020 Year 11 Examination

### Mathematics Standard

#### Task 3: Examination

##### General

##### Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black pen
- NESA approved calculators may be used
- A reference sheet is provided at the back of this paper
- For questions in Section II, show relevant mathematical reasoning and/or calculations

**Total marks:**  
**80**

##### Section I – 15 marks (pages 2–6)

- Attempt Questions 1–15
- Write your answers on the answer sheet
- Allow about 25 minutes for this section

##### Section II – 65 marks (pages 7–20)

- Attempt Questions 16–19
- Allow about 1 hour and 35 minutes for this section

#### To be completed during the marking and checking process

| Question        | Multiple Choice | 16 | 17 | 18 | 19 | Total |
|-----------------|-----------------|----|----|----|----|-------|
| Marks           | 15              | 16 | 18 | 16 | 15 | 80    |
| Student's Score |                 |    |    |    |    |       |

Student's comment:

Teacher's comment:

#### Disclaimer

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MSYR11\_20\_EXAM

# Section I

15 marks

Attempt Questions 1–15

Allow about 25 minutes for this section

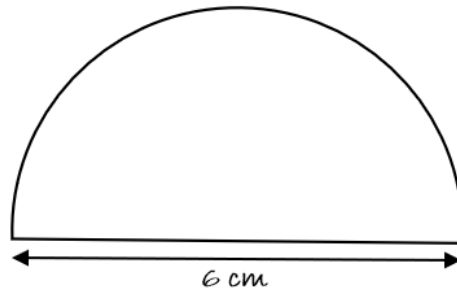
Use the multiple-choice answer sheet for Questions 1–15.

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- 1 Calculate the value of  $\frac{3x-y}{3}$  if  $x = 50$  and  $y = 24$ .
- A. 26  
B. 42  
C. 142  
D. 342
- 2 Taylah invests \$8000 in an account with simple interest at a rate of 3.5% p.a. for 4 years. How much is the investment worth at the end of the 4-year term?
- A. \$280  
B. \$1120  
C. \$8280  
D. \$9120
- 3 Fried's formula, shown below, is used to calculate the dosage of medication for children aged 1-2 years.
- Fried's formula (children 1-2 years of age):  $Dosage = \frac{(age\ in\ months) \times adult\ dose}{150}$
- Lola is  $1\frac{1}{2}$  years old and requires a dose of medicine which has an adult dosage of 500 mg. Calculate Lola's required dosage.
- A. 3.5 mg.  
B. 5 mg.  
C. 35 mg.  
D. 60 mg.

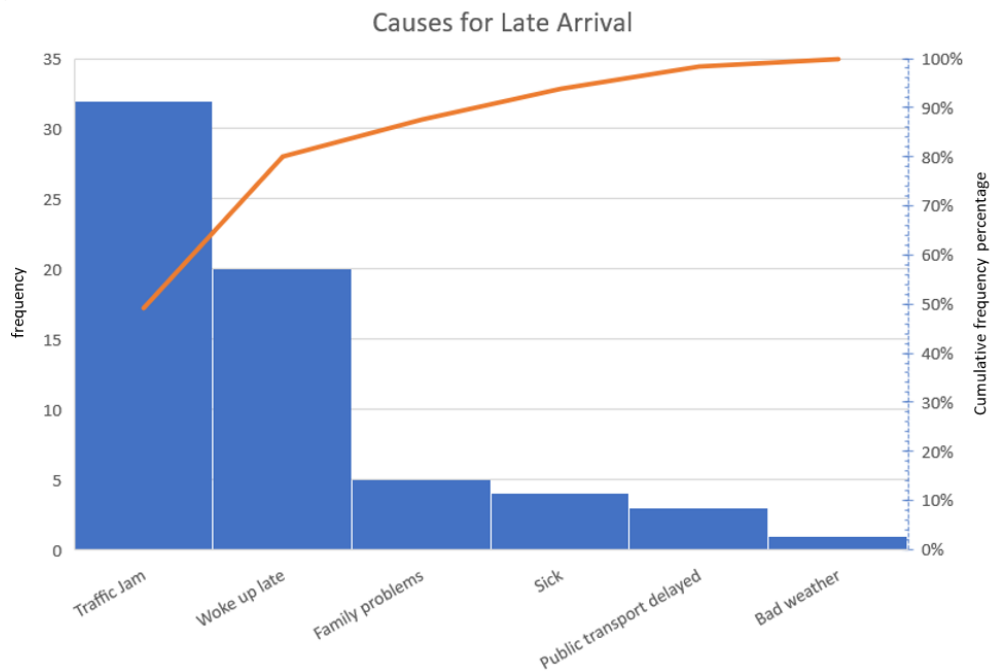
- 4 Sharon's business sells T-shirts in a range of sizes, (XS, S, M, L, XL). To manage her stock levels Sharon must keep track of how many of each size are sold. What type of data is Sharon gathering?
- A. Categorical Ordinal
  - B. Categorical Nominal
  - C. Numerical Discrete
  - D. Numerical Continuous
- 5 Express 0.000 005 25 in Scientific notation, correct to 2 significant figures.
- A. 0.0
  - B.  $5.25 \times 10^{-5}$
  - C.  $5.25 \times 10^{-6}$
  - D.  $5.3 \times 10^{-6}$
- 6 Tani wishes to call her grandmother for her birthday. Tani lives in Perth (UTC +8) and her grandmother lives in Auckland (UTC +12). What time in Perth will Tani need to call if she wants it to be 11am in Auckland?
- A. 7 am
  - B. 3 pm
  - C. 7 pm
  - D. 11 pm
- 7 In a "15% Off Sale" Pim saves \$19.50 off the price of a new hockey stick. How much did she pay for the hockey stick?
- A. \$16.58
  - B. \$110.50
  - C. \$130.00
  - D. \$149.50
- 8 What is the value of  $x$  in the equation  $2(5 - x) = 6$ ?
- A. 2
  - B. -2
  - C. 7
  - D. -17

- 9 To the nearest square centimetre, what is the area of the semi-circle shown below?



- A.  $14 \text{ cm}^2$   
B.  $28 \text{ cm}^2$   
C.  $57 \text{ cm}^2$   
D.  $113 \text{ cm}^2$
- 10 A damaged water tank is leaking at a rate of 3L/hour.  
If it initially contained 600L, which of the following represents the volume of water remaining in the tank (V) after (t) hours?
- A.  $V = 600t - 3$   
B.  $V = 3 - 600t$   
C.  $V = 600 + 3t$   
D.  $V = 600 - 3t$
- 11 The capacity of a motorcycle's fuel tank is measured as 12.4L.  
What is the absolute error of this measurement?
- A. 10 ml  
B. 50 ml  
C. 100 ml  
D. 500 ml
- 12 If  $c = -2$ , find the value of  $\frac{3}{c^2} - \frac{2}{c^3}$
- A.  $\frac{1}{2}$   
B.  $-\frac{1}{2}$   
C. 1  
D. -1

13 Consider the Pareto chart below.



What percentage of late arrivals were due to waking up late?

- A. 20%
- B. 31%
- C. 52%
- D. 80%

14 Demetrios depreciates his office computer equipment using the straight line method. After depreciating for 3 years at 18% p.a., the equipment has lost \$1971 of its initial value.  
How much is its current salvage value?

- A. \$1064.34
- B. \$1679
- C. \$3650
- D. \$5913

- 15 The five-figure summary of a data set is 9, 20, 24, 27, 38.  
This data set will have:
- A. no outliers.
  - B. at least 1 lower outlier.
  - C. at least one upper outlier.
  - D. both upper and lower outliers.

**End of Section I**

# Mathematics Standard

## Section II Answer Booklet

**65 marks**

**Attempt Questions 16–19**

**Allow about 1 hours and 35 minutes for this section**

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**Instructions**

- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
  - Your responses should include relevant mathematical reasoning and/or calculations.
  - Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.
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**Question 16** (16 marks)

- (a) Simplify

$$5(2m - 4) - 3(2m - 5)$$

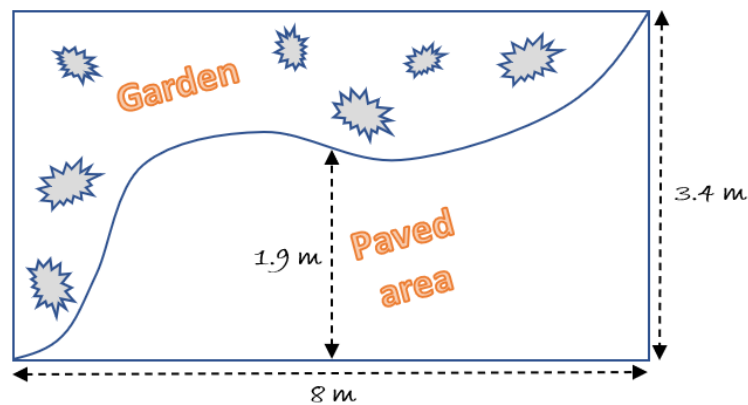
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- (b) The rectangular courtyard shown is split into a garden bed and a paved area. In the diagram the paved area has been divided into two strips of equal width.



Use 2 applications of the Trapezoidal Rule to approximate the area of the paved area.

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- (c) Given the formula  $r = \sqrt{\frac{A}{4\pi}}$

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Find the value of  $r$ , to 2 significant figures, when  $A = 16.4$ .

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Question 16 (continued)

- (d) Jacob is paid 4% commission on his total sales over the first \$10 000.  
Calculate his total sales figure in a week where his commission was \$2200. 2

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- (e) A stratified sample of 55 students is to be surveyed of the Year 11 and 12 students from Green Hills High School.  
The student numbers for each year group are as follows:  
150 students in Year 11 (80 boys and 70 girls)  
125 students in Year 12 (65 boys and 60 girls)  
How many of the Year 11 girls should be included in the survey? 1

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- (f) Paris lies on the  $2^{\circ}E$  line of longitude and New York lies on  $76^{\circ}W$  line.  
Calculate the time difference between the cities. 2

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- (g) Petra runs on her treadmill at a speed of 18km/hr.  
What is this speed in m/sec? 2

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Question 16 (continued)

- (h) A muesli bar contains 6.7 grams of fat.  
Assuming that 1 gram of fat contains 17 kilojoules (kJ), use the conversion  
 $1 \text{ Calorie} = 4.184 \text{ kJ}$  to determine how many calories in the muesli bar are  
due to its fat content. Answer correct to 3 decimal places.

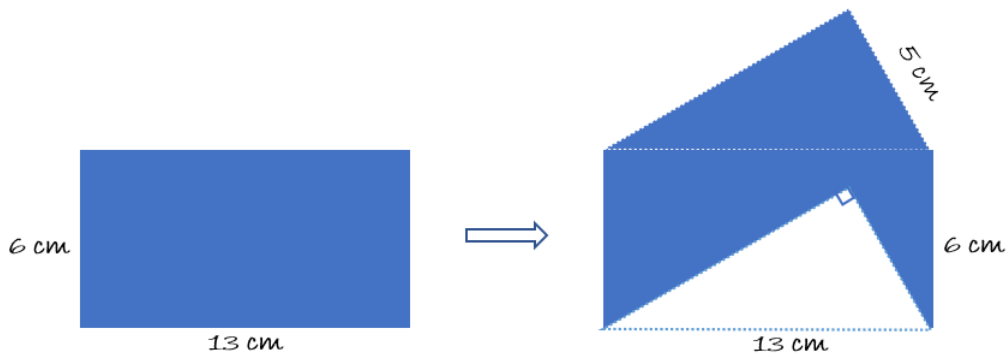
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**End of Question 16**

**Question 17** (18 marks)

- (a) A  $13 \times 6$  cm rectangle has a right-angled triangle removed from one edge which is then attached to the opposite edge, forming the composite shape below.



- (i) What is the area of the composite shape formed? 1

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- (ii) What is the perimeter of the composite shape formed? 2

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- (b) Savvas earns \$96 000 as an annual gross salary.  
 When taking his 4 weeks annual leave, he is paid 17.5% leave loading in addition to his normal pay.  
 How much will Savvas receive for his 4 weeks annual leave? 2

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Question 17 (continued)

- (c) In a household of 5 people, each person in the house uses an average of 11 minutes of hot water every day. The house has a 9.9 kW hot water system. The cost of electricity is 27.56 cents/kWh. At this rate, how much does the household pay for electricity for the hot water system in one year?

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- (d) An experiment to test the accuracy of a lie detector machine was conducted. 150 people were tested, and their results were recorded below.

- 30 people LIED, but the test only indicated 24 had lied.
- Of the people who did NOT LIE, the test indicated 18 had lied.

- (i) Using the results above, complete the two-way table below.

1

|                        | Test indicated LIE | Test did not indicate a LIE | Total |
|------------------------|--------------------|-----------------------------|-------|
| People who LIED        | 24                 |                             | 30    |
| People who did NOT LIE | 18                 |                             | 120   |
| Total                  | 42                 | 108                         | 150   |

- (ii) What percentage of the test results was accurate?

2

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**Question 17 continues on next page**

Question 17 (continued)

- (e) The diagram below models the additional costs for extra data usage on two mobile phone plans.



- (i) Rachel is on Plan B and needs an extra 200 Gigabytes. How much extra will she need to pay?

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- (ii) Rohan has \$50 to spend on additional data. Which plan should he join to get the most data for his money? How much extra data would he get when compared to the other plan?

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- (iii) What is the gradient of the Plan B line? What does this represent?

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**Question 17 continues on next page**

- (f) Alberto lives in Sydney (UTC +10) and wants to stream a live basketball match being played in Chicago (UTC -6).  
The game starts 8pm Thursday in Chicago, what time (and day) is it in Sydney when the game starts?

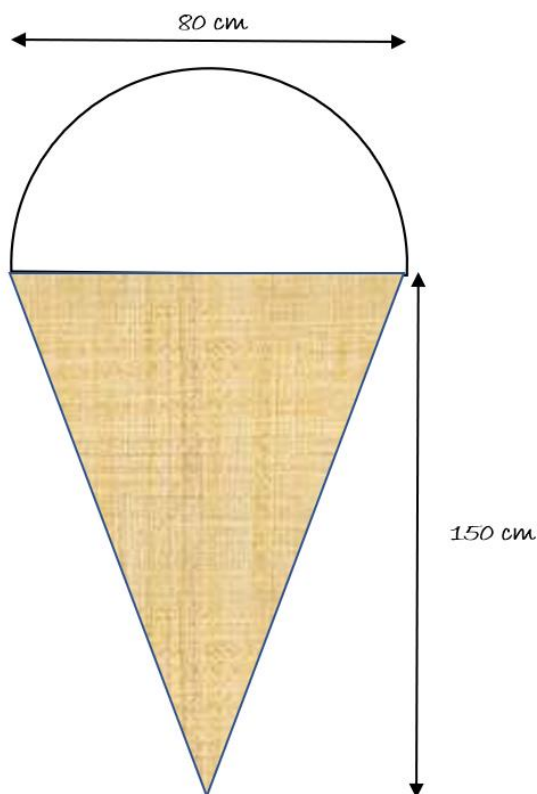
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**End of Question 17**

**Question 18** (16 marks)

- (a) Isabelle's Ice Creams are getting a large plastic model made for promotional purposes. As shown below, the model consists of a hemisphere, with diameter of 80 cm, joined to a 150 cm tall cone.



- (i) How tall will the model be when completed? **1**

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- (ii) Calculate the volume of plastic required to make the model. **3**  
Give your answer to the nearest cubic centimetre.

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**Question 18 continues on next page**

Question 18 (continued)

- (b) Last year Analise's taxable income was \$98 750.

| <i>Taxable income</i> | <i>Tax payable on this income</i>             |
|-----------------------|-----------------------------------------------|
| \$0 to \$18 200       | Nil                                           |
| \$18 201 to \$37 000  | 19c for each \$1 over \$18 200                |
| \$37 001 to \$90 000  | \$3 572 plus 32.5c for each \$1 over \$37 000 |
| \$90 001 to \$180 000 | \$20 797 plus 37c for each \$1 over \$90 000  |
| \$180 001 and over    | \$54 097 plus 45c for each \$1 over \$180 000 |

- (i) Calculate her tax payable using the table above. 1

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- (ii) In addition to tax, Analise must pay the Medicare levy which is set at 2% of her taxable income. How much Medicare levy does Analise owe? 1

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- (iii) During the year, Analise has already paid \$28080 in PAYG tax instalments to cover her income tax and Medicare levy. After completing her tax return, will Analise receive a tax refund or owe more in tax? Justify your answer with supporting calculations. 2

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**Question 18 continues on next page**

Question 18 (continued)

- (c) Solve for  $x$ .

$$3(x - 10) = 3 - \frac{2x}{3}$$

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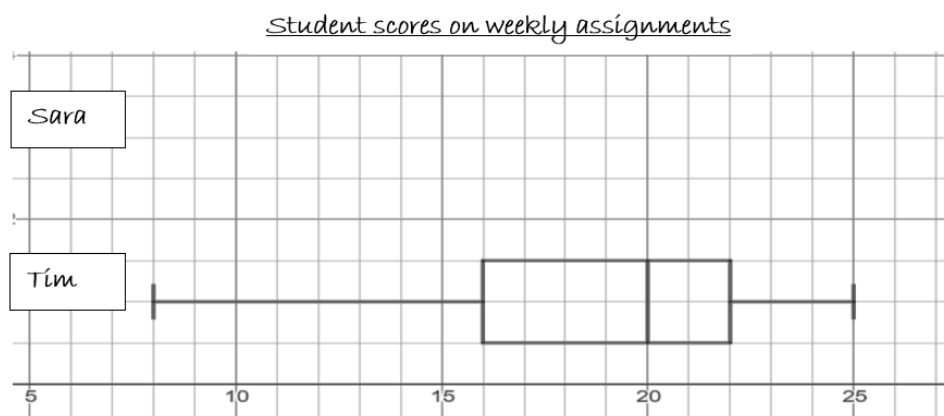
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- (d) Tim and Sara are comparing their results from weekly class assignments. They each generate a five-figure summary of their results and Tim draws the box plot below to represent his scores.



- (i) Sara's five figure summary is 10, 12, 15, 20, 26.  
Draw her box plot above Tim's.

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- (ii) Who had the better results overall?  
Justify your answer by referring to the summary statistics and the shapes of the distributions.

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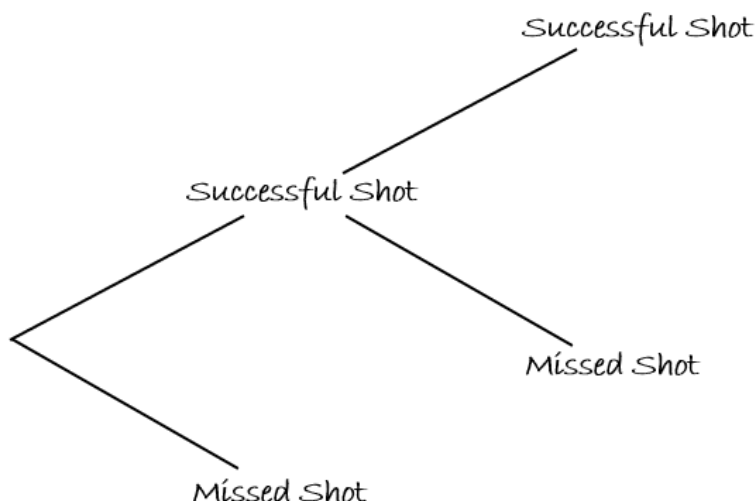
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**End of Question 18**

**Question 19** (15 marks)

- (a) During a game of basketball, Mia is awarded a penalty. She can have two shots at the basket, but if the first is not successful she cannot take the second shot. Each successful shot is worth 1 point. The probability of success for each of her shots is 0.6.



- (i) Complete the probability tree by writing the probabilities on all the branches. 1

- (ii) List the outcomes shown on the probability tree using S for success and M for miss and their corresponding point scores. 2

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- (iii) What is the probability of Mia scoring 2 points? 1

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- (iv) What is the most likely score? 1

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**Question 19 continues on next page**

Question 19 (continued)

- (b) Rewrite the formula to make  $A$  the subject.

$$P = Ae^{mt} + 1000$$

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- (c) Donald was given the set of scores: 58, 62, 43, 54, 68, 64, 58, 56, 49, 92. He analysed the scores by finding the mean, standard deviation,  $Q_1$  and  $Q_3$ . Donald concluded that 92 was an outlier.

- (i) Verify Donald's conclusion about the outlier by completing your own analysis .

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- (ii) Remove the outlier and redo the analysis of the data. What effect did the outlier have on the statistical analysis?

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**Question 19 continues on next page**

Question 19 (continued)

- (d) Pete's Pizza can make a traditional pizza for a cost of  $\$C$ .  
To upgrade the pizza to a seafood pizza, Pete calculates that the cost increases to  $\$\frac{4C}{3}$ .  
Pete makes a profit of 50% on all pizzas sold.  
A customer orders one traditional pizza and one seafood pizza. Show that the selling price for the order is  $\$\frac{7C}{2}$ .

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**End of Question 19**

**End of Section II**

**End of Exam**

## Section II extra writing space

If you use this space, clearly indicate which question you are answering.

Student Name:.....

**MATHEMATICS STANDARD – MULTIPLE-CHOICE ANSWER SHEET**

ATTEMPT ALL QUESTIONS

| Question | 1  | A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |
|----------|----|-------------------------|-------------------------|-------------------------|-------------------------|
|          | 2  | A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |
|          | 3  | A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |
|          | 4  | A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |
|          | 5  | A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |
|          | 6  | A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |
|          | 7  | A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |
|          | 8  | A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |
|          | 9  | A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |
|          | 10 | A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |
|          | 11 | A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |
|          | 12 | A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |
|          | 13 | A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |
|          | 14 | A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |
|          | 15 | A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |

# Mathematics Standard Reference Sheet

## Measurement

### Precision

$$\text{Absolute error} = \frac{1}{2} \times \text{precision}$$

$$\text{Upper bound} = \text{measurement} + \text{absolute error}$$

$$\text{Lower bound} = \text{measurement} - \text{absolute error}$$

### Length, area, surface area and volume

$$l = \frac{\theta}{360} \times 2\pi r$$

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2} (x + y)$$

$$A = \frac{h}{2} (d_f + d_l)$$

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$

### Trigonometry

$$A = \frac{1}{2}ab \sin C$$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

## Financial Mathematics

$$FV = PV(1 + r)^n$$

### Straight-line method of depreciation

$$S = V_0 - Dn$$

### Declining-balance method of depreciation

$$S = V_0(1 - r)^n$$

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## Statistical Analysis

$$z = \frac{x - \bar{x}}{s}$$

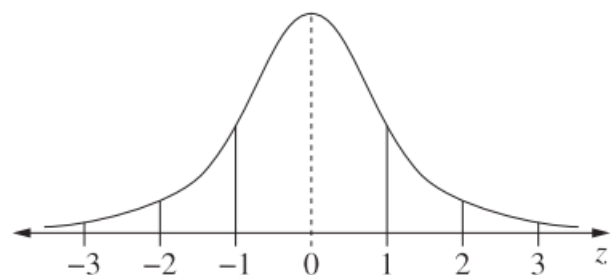
An outlier is a score

$$\text{less than } Q_1 - 1.5 \times IQR$$

or

$$\text{more than } Q_3 + 1.5 \times IQR$$

### Normal distribution



- approximately 68% of scores have z-scores between  $-1$  and  $1$
- approximately 95% of scores have z-scores between  $-2$  and  $2$
- approximately 99.7% of scores have z-scores between  $-3$  and  $3$